

PD17 — Empirical MBB characterization

Knobs (mini glossary)

Symbol	Plain name	What it controls
$\hat{A}_i$	Player ability	PPM within season
$\hat{T}$	Realized team talent	Mean $\hat{A}_i$ on roster, not T_{j*}
T_{j*}	Sim assignment target	Pass B only - not observed in NCAA data
ρ	Assignment assortativity	Sim soft-match knob; overlay forensics on data
L_C	Team congestion	$mean_i \sigma(\gamma(\hat{A}_i - \theta))$ per roster
λ	Congestion weight	Score $S_i = A_i - \lambda \cdot L_C$ (sim / later on data)
θ	Viability cutline	$F_{\hat{A}}^{-1}(1 - K/N)$ on empirical $\hat{A}_i$
γ	Sigmoid sharpness	Reshapes team L_C - 539 placeholder until lock
K/N	Draft rate	Accepted / total in filtered panel ($\approx .1.8\%$)

Deck map (assign → score → comparability → Hero)

- Block A — Empirical inputs & overlap**
 - 2 · $\hat{A}_i, \hat{T}_j$ inputs (2011–2021)
 - 3 · NCAA interval overlap (full panel)
 - 4–5 · NCAA vs LG intervals (2015–2019 window)
- Block B — LG ρ validation (ASSIGN)**
 - 6 · global_wss vs ρ (primary)
 - 7 · H_{sort} glossary + ρ inset
- Block C — Team L_C on real rosters (SCORE input)**
 - 8 · Empirical team L_C ($\gamma = 0.5$)
- Block D — LG input comparability (empirical caps, $\rho = 0.5$)**
 - 9 · Roster-size multiset (Alex caps fix)
 - 10 · Team L_C vs NCAA (6,492 team-seasons)
 - 11 · SELECT vs Hero — caps alone $\neq$ inverted-U $\rightarrow \lambda$ sweep
- Block E — Hero & sensitivity (empirical)**
 - 12 · NCAA roster sizes vs fixed $C = 15$
 - 13–14 · min_minutes ladder (Hero robustness)
 - 15–16 · LG C sweep (L_C + SELECT; monotone)
- Block F — λ in SCORE (LG, empirical caps held fixed)**
 - 20 · λ fine grid (flip 1.25 $\rightarrow$ 1.5; peak bin 11 at $\lambda \approx 2-3$)
 - 21 · λ extended grid
- Block G — Empirical SCORE diagnostics (PD17)**
 - 22 · Sketch A: T_j vs L_C
 - 23 · γ sweep on real rosters

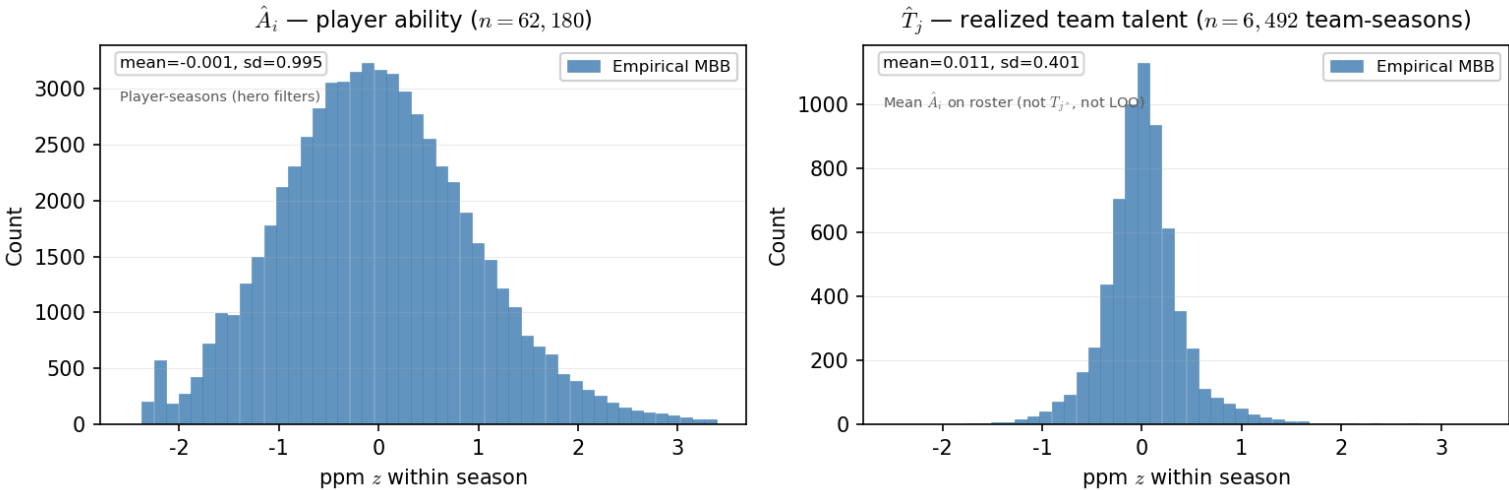
Claim (PD17): Map empirical roster geometry; validate ρ on ASSIGN; with caps fixed, **Hero shape on LOO emerges in LG via λ in SCORE** (slides 20–21) — not via ρ or roster alignment alone.

PD17 — Empirical MBB inputs: $\hat{A}_i$ and $\hat{T}_j$

MBB 2011-2021 · ppm z within season · min 20 min · poolq winsor 0.01–0.99

- Realized rosters — descriptive inputs, not sim draws.
- Left: $\hat{A}_i$ — player ability (one value per player-season).
- Right: $\hat{T}_j$ — mean $\hat{A}_i$ on roster (team-season; not LOO).
- $\hat{T}_j$ is tighter than $\hat{A}_i$ (averaging shrinks spread).
- This run: mean $\hat{A}_i$ =-0.001, mean $\hat{T}_j$ =0.011 (sd 0.995 / 0.401).
- **PD17 θ proxy: K/N = 0.0182.**
(1,134 drafted / 62,180 player-seasons).

Empirical MBB inputs — $\hat{A}_i$ and $\hat{T}_j$ (2011–2021, min 20 min, poolq winsor 0.01–0.99)



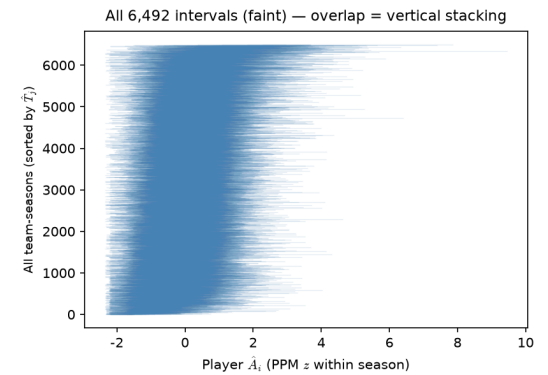
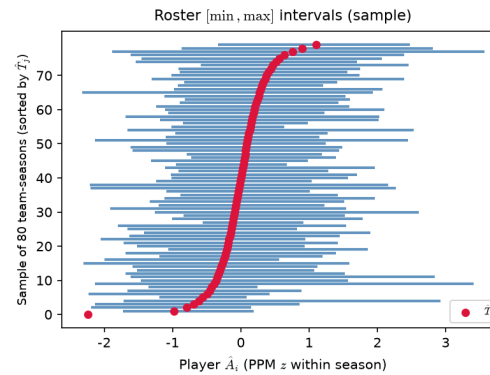
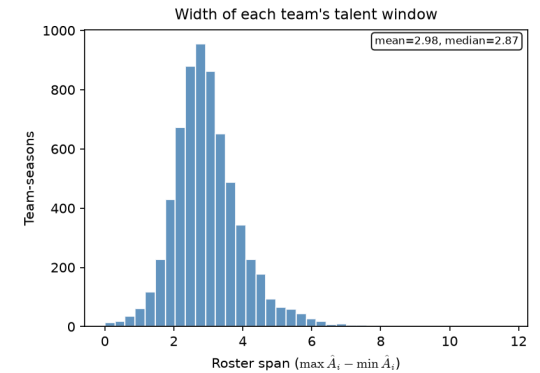
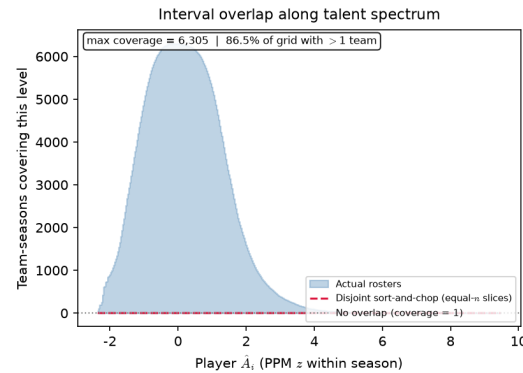
Claim (PD17): World-first inputs from the real MBB panel — $\hat{A}_i$ per player-season and $\hat{T}_j$ per team-season. Not T_j^* (sim assignment targets).

PD17 — Empirical team $\hat{A}_i$ interval overlap (ρ diagnostic)

MBB 2011-2021 · PPM z within season · min 20 min · poolq winsor 0.01–0.99 · 6,492 team-seasons (530 CELL 8 port) · $H_{sort} = 0.105$

Empirical NCAA — team talent window overlap (MBB 2011-2021)\nRealized sorting $H_{sort} = 0.105$ on this partition

- Each team-season: talent window $[\min \hat{A}_i, \max \hat{A}_i]$ on PPM z.
- Coverage = how many windows cover a point on the spectrum (530 CELL 8).
- Red dashed: disjoint sort-and-chop on same player-seasons (537 B analog).
- $H_{sort} = 1 - \sum_i (\hat{A}_i - \hat{\mu}_{g(i)})^2 / \sum_i (\hat{A}_i - A)^2$ — realized sorting on a fixed partition.
- NCAA partition: $H_{sort} = 0.105$ (0 = none, 1 = full team sorting).
- Max coverage = 6,305; sort-and-chop max = 7. Normalized peak = 0.97.
- 86% of grid points have > 1 team covering.
- Roster span: mean = 2.98 z, median = 2.87 z (6,492 team-seasons).
- Full panel 2011–2021 — pooled team-seasons (not 1:1 with single-season sim).
- VECTOR lock: H_{sort} is realized sorting — not the generative homophily knob ρ .



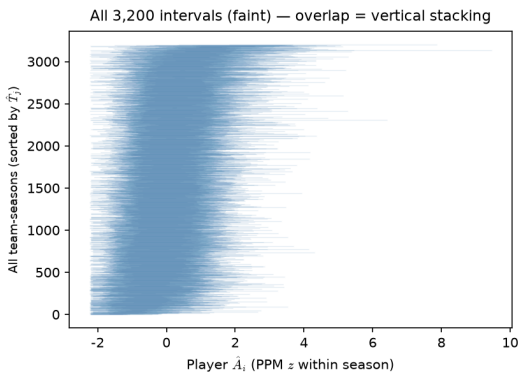
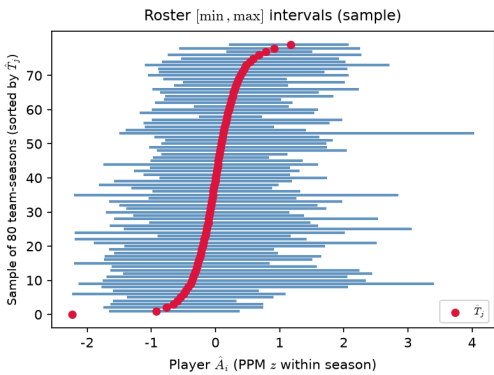
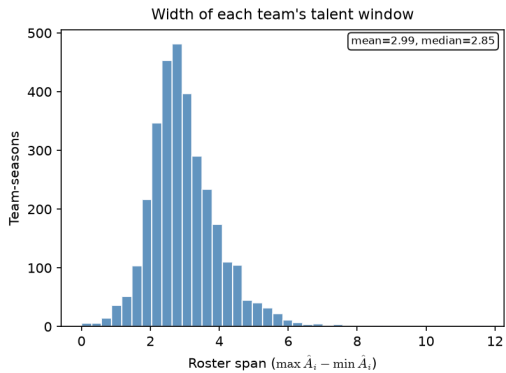
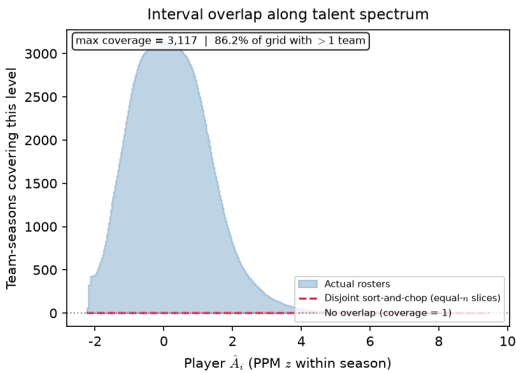
Claim (PD17): Real NCAA rosters leave massively overlapping team talent windows — full 2011–2021 panel target for ρ characterization (530 CELL 8).

PD17 — NCAA team $\hat{A}_i$ interval overlap (2015-2019)

MBB 2015-2019 · PPM z within season · min 20 min · poolq winsor 0.01–0.99 · 3,200 team-seasons (530 CELL 8 port) · $H_{sort} = 0.105$

Empirical NCAA — team talent window overlap (MBB 2015-2019)\nRealized sorting $H_{sort} = 0.105$ on this partition

- Each team-season: talent window $[\min \hat{A}_i, \max \hat{A}_i]$ on PPM z.
- Coverage = how many windows cover a point on the spectrum (530 CELL 8).
- Red dashed: disjoint sort-and-chop on same player-seasons (537 B analog).
- $H_{sort} = 1 - \sum_i (\hat{A}_i - \hat{\mu}_{g(i)})^2 / \sum_i (\hat{A}_i - A)^2$ — realized sorting on a fixed partition.
- NCAA partition: $H_{sort} = 0.105$ (0 = none, 1 = full team sorting).
- Max coverage=3,117; sort-and-chop max=7. Normalized peak=0.97.
- 86% of grid points have > 1 team covering.
- Roster span: mean=2.99 z, median=2.85 z (3,200 team-seasons).
- Paired compare window — match season span to LG sim slide.
- VECTOR lock: H_{sort} is realized sorting — not the generative homophily knob ρ .



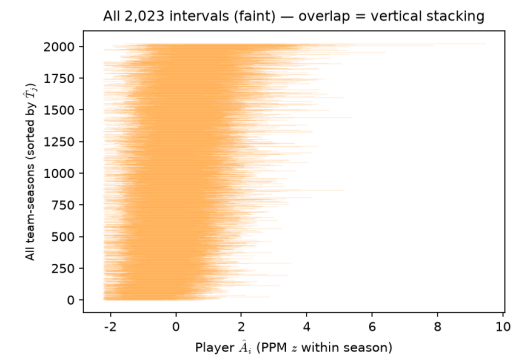
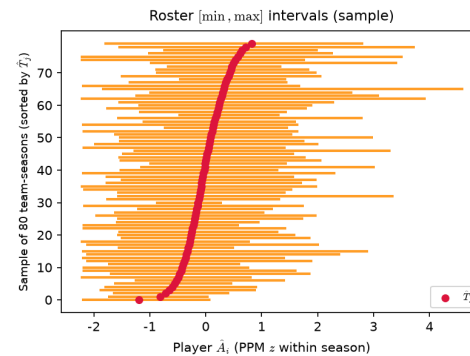
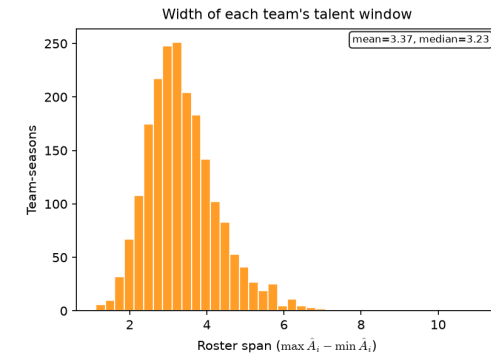
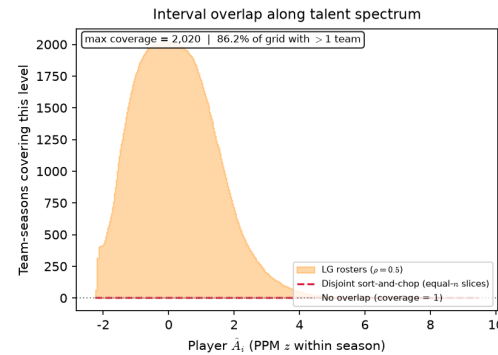
Claim: NCAA interval overlap on a fixed season window — paired compare to LG ASSIGN sim on the same window.

LG sim — team $\hat{A}_i$ interval overlap (2015-2019)

MBB 2015-2019 · LG ASSIGN · $\rho = 0.5$ · $C = 15$ · $N = 30,345$ 2,023 team-seasons · $H_{sort} = 0.146$

- Each team-season: talent window $[\min \hat{A}_i, \max \hat{A}_i]$ [on PPM z.
- Coverage = how many windows cover a point on the spectrum (530 CELL 8).
- LG ASSIGN: $\rho = 0.5$, endogenous μ_j — no T_j^* .
- Red dashed: disjoint sort-and-chop on same player pool (537 B analog).
- $H_{sort} = 1 - \sum_i (\hat{A}_i - \hat{\mu}_{g(i)})^2 / \sum_i (\hat{A}_i - A)^2$ — realized sorting on a fixed partition.
- Assign partition: $H_{\{sort\}} = 0.146$ (0 = none, 1 = full team sorting).
- Sim max coverage= 2,020; sort-and-chop max= 5. Normalized peak= 1.00.
- 86% of grid points have > 1 team covering.
- Roster span: mean =3.37 z, median =3.23 z (2,023 team-seasons).
- NCAA ref max cov= 3,117; NCAA overlap grid= 86%; NCAA $H_{sort} = 0.105$.
- VECTOR lock: H_{sort} is realized sorting — not the generative homophily knob ρ .

LG ASSIGN — team talent window overlap (MBB 2015-2019, $\rho = 0.5$, $C = 15$)
Realized sorting $H_{sort} = 0.146$, global_wss = 25,881 on this partition

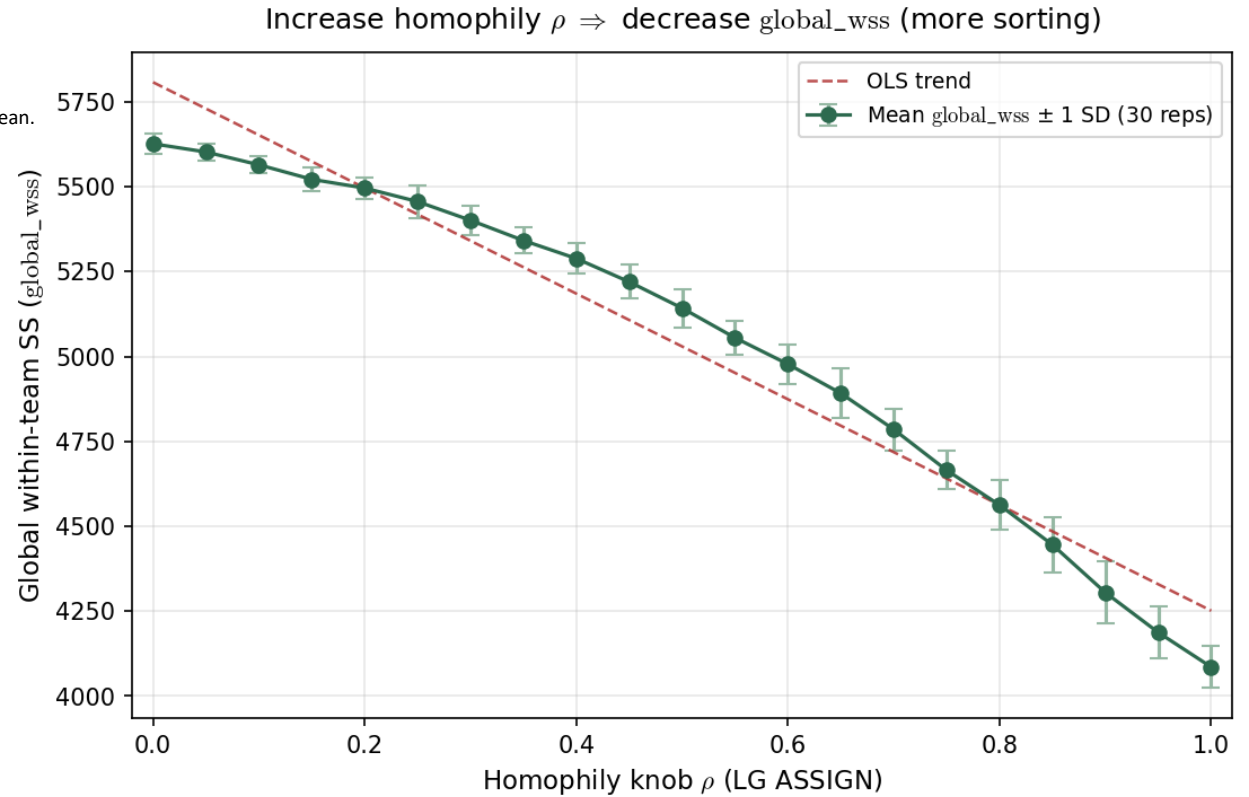


Claim: LG ASSIGN on the same season window as paired NCAA slide — one realization per season, stacked team-seasons.

LG ASSIGN — $global_wss$ vs ρ — total global within-team sum of squared-errors

LG ASSIGN · 2015 PPM $z \cdot J = 402, C = 15 \cdot \rho \in [0,1] \cdot 30$ reps/arm

- ρ = LG ASSIGN homophily knob in $\exp(-\rho|\hat{A}_i - \mu_j|)$.
- Each point: mean ± 1 SD of $global_wss$ over stochastic LG realizations.
- $global_wss = \sum_j \sum_{i \in j} (\hat{A}_i - \hat{T}_j)^2$ — global within-team SS; $\hat{T}_j$ = realized roster mean.
- $H_{sort} = 1 - global_wss/SS_{total}$ on the same partition
 $global_wss$ is raw magnitude;
 H_{sort} is scale-free explained variance.
- $\rho = 0$: $global_wss \approx 5627 \rightarrow \rho = 1$: $global_wss \approx 4085$.
- 2015 empirical PPM $z \cdot J = 402 \cdot C = 15 \cdot 30$ reps $\times 21 \rho$ arms.
- $\rho \uparrow$ sorting $\Rightarrow global_wss \downarrow, H_{sort} \uparrow$ (monotone mirror on a fixed ability pool).
- Companion slide: $H_{sort} = 1 - global_wss/SS_{total}$
(scale-free assortativity on the same sweep).
- VECTOR lock: H_{sort} is realized sorting — not the generative homophily knob ρ .



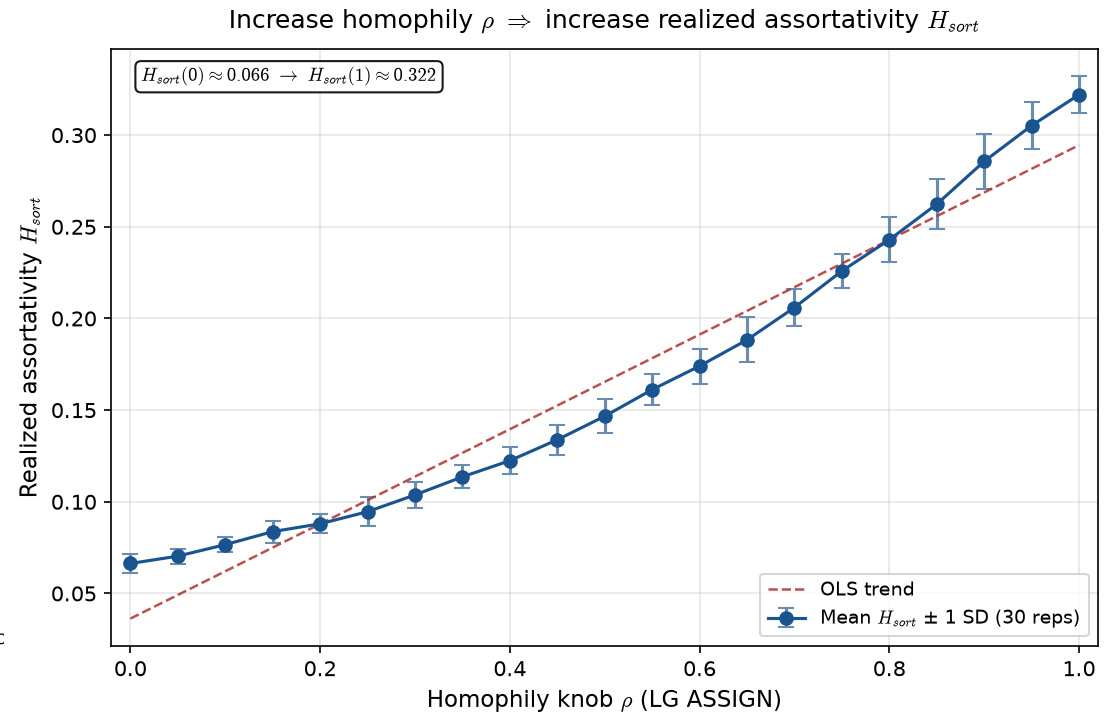
Claim (Alex): $global_wss$ is the within-team SS numerator on the assign partition — $\rho \uparrow$ lowers $global_wss$ on the same 2015 LG league (H_{sort} is the scale-free companion).

What is H_{sort} ? Realized assortativity on rosters

Realized sorting index · LG / PD17 · alias `sorting_index_h` in code

$$H_{sort} = 1 - \frac{\sum_i (\hat{A}_i - \hat{\mu}_{g(i)})^2}{\sum_i (\hat{A}_i - \bar{A})^2} - \text{realized sorting on a fixed partition.}$$

- Denominator: total spread of $\hat{A}_i$ around league mean A .
- Numerator: spread of $\hat{A}_i$ around own team mean $\mu_{g(i)}$.
- $H_{sort} = 1 - (\text{within-team SS}) / (\text{total SS})$ — explained-variance / ANOVA-style.
- $H_{sort} \approx 0$: team label tells you little (random-like partition).
- $H_{sort} \approx 1$: teams are perfectly separated (point-mass rosters).
- Alex one-liner: knowing roster explains how much ability deviates from A .
- Not Newman network assortativity r — partition statistic on abilities.
- Complements interval overlap (geometry) — use both in brief.
- $\rho \uparrow \Rightarrow H_{sort} \uparrow$ in LG sweep (see inset plot).
- VECTOR lock: H_{sort} is realized sorting — not the generative homophily knob ρ .
- Memo: `grandchild_assign`/`GRANDCHILD_D_and_H_sort_interpretatic`



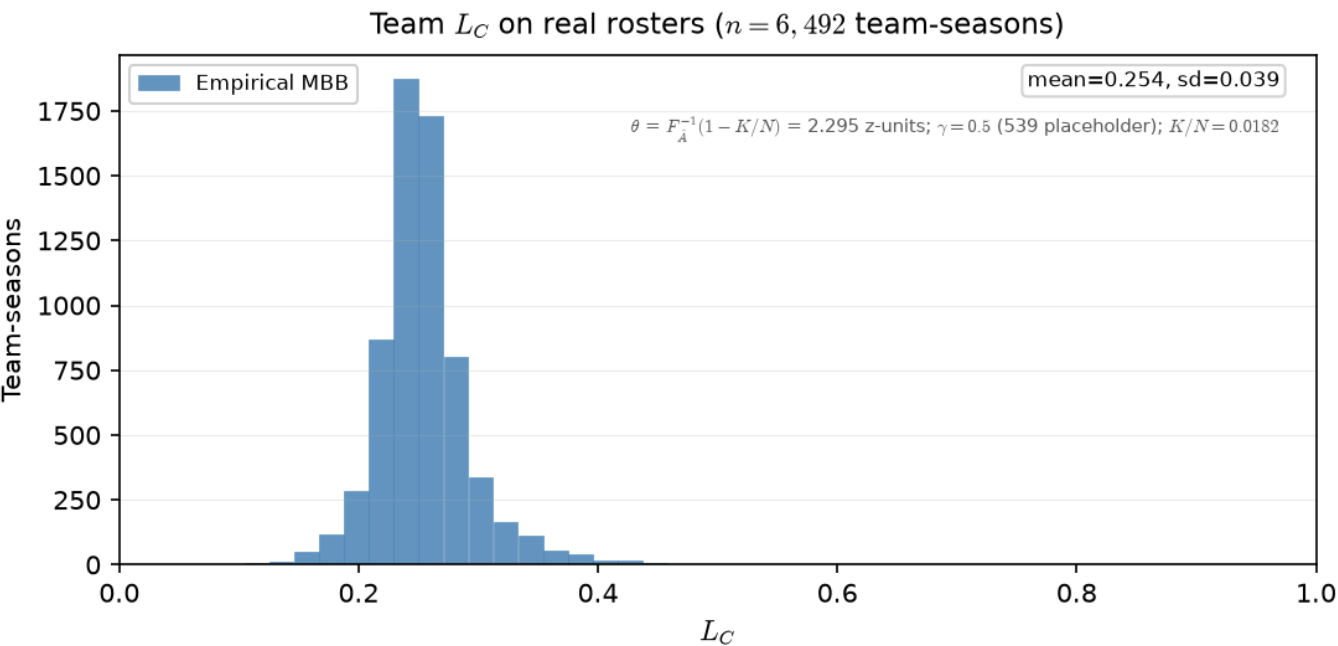
Claim: H_{sort} is realized assortativity on a fixed partition — generative homophily ρ is the ASSIGN knob; measure H_{sort} after assign.

PD17 — Empirical team L_C distribution

MBB 2011-2021 · team smooth L_C · PPM z within season · min 20 min · $\theta = F_A^{-1}(1 - K/N) = 2.295$ ($K/N = 0.0182$) · $\gamma = 0.5$ placeholder

- Team $L_C = \text{mean}_j \sigma(\gamma(\hat{A}_j - \theta))$ on each roster.
- Same PD16 team-smooth definition — one value per team-season.
- Not LOO; includes every roster member in the mean.
- $\gamma = 0.5$ (539 placeholder until PD17 γ / λ calibration).
- This run: mean $L_C = 0.254$, sd = 0.039 (6,492 team-seasons).
- θ from draft rate: $K/N = 0.0182 \Rightarrow \theta = 2.295$ on $\hat{A}_i$.

Empirical MBB — team smooth L_C (2011-2021, min 20 min, poolq winsor 0.01-0.99)



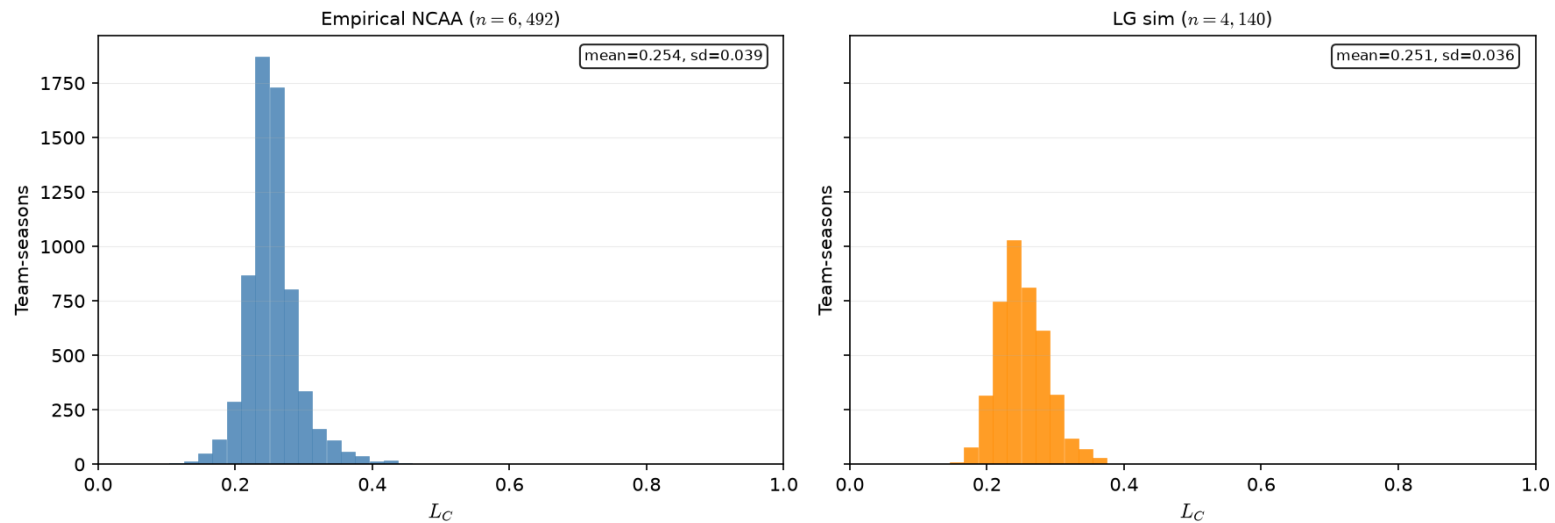
Claim (PD17): Real rosters carry a spread of team congestion L_C — the descriptive input ρ was meant to stratify in sim.

Empirical vs LG — team L_C distribution vs NCAA

MBB 2011-2021 · 6,492 team-seasons · NCAA mean=0.254

- Team L_C after empirical-cap ASSIGN vs real NCAA rosters.
- MBB 2011-2021 · $\rho = 0.5$, $\gamma = 0.5$.
- NCAA: $n = 6,492$ · mean = 0.254 · sd = 0.039.
- LG empirical caps: $n = 6,492$ · mean = 0.251 · sd = 0.045.
- Team count now aligned (6,492) — fixed $C = 15$ had only 4,140 teams.
- L_C first moment still matches; slightly wider LG tail (assign noise).
- Input comparability fixed; SCORE/SELECT shape is a separate question.

Team smooth L_C — empirical vs LG (MBB 2011-2021, $\gamma = 0.5$, $\theta = 2.295$ z, $K/N = 0.0182$)



Talking point: comparability achieved — same team count and L_C mean; assignment layer is now apples-to-apples on roster geometry.

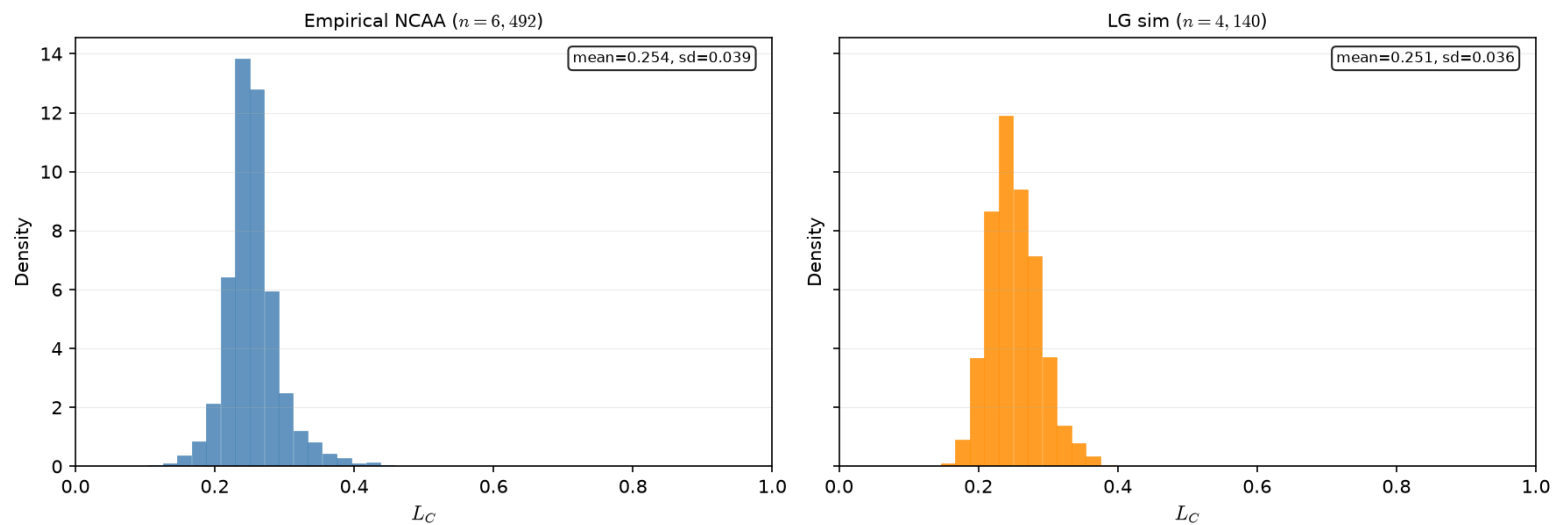
Empirical vs LG — team L_C distribution (normalized)

MBB 2011-2021 · team smooth L_C · LG $\rho = 0.5$ · $\gamma = 0.5$ · $\theta = 2.295$ z ($K/N = 0.0182$)

- Team $L_C = \text{mean}_j \sigma(\gamma(\hat{A}_j - \theta))$ — team-smooth congestion.
- Left panel: empirical NCAA (real rosters). Right: LG ASSIGN sim.
- Panel: MBB 2011-2021 · PPM z · min 20 min · poolq winsor 0.01–0.99.
- Shared $\theta = F_A^{-1}(1 - K/N) = 2.295$ z ($K/N = 0.0182$).
- $\gamma = 0.5$ (HAND17 slide-4 default; 539 placeholder).
- Empirical: mean $L_C = 0.254$, sd= 0.039 (6,492 team-seasons).
- LG ($\rho = 0.5$): mean $L_C = 0.251$, sd= 0.036 (4,140 team-seasons).
- Sim: one LG league per season ($J = N/15$), stacked team-seasons.
- ASSIGN shapes rosters; L_C is score-side input — computed after assign.

Same L_C formula; normalized so team-season count differences don't dominate the visual.

Team smooth L_C — normalized compare (MBB 2011-2021, $\gamma = 0.5$, $\theta = 2.295$ z, $K/N = 0.0182$)



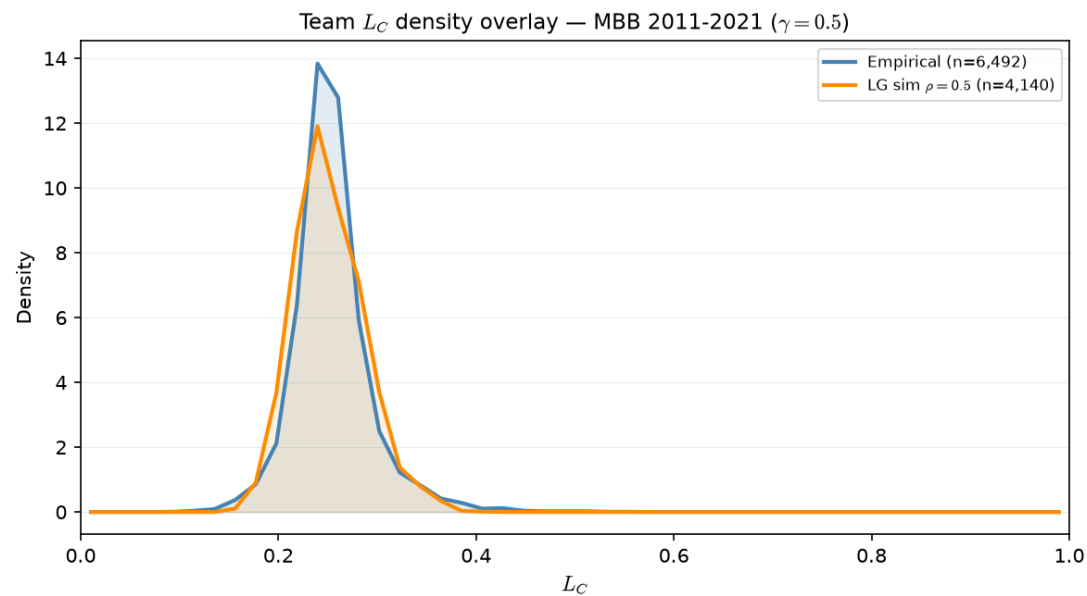
Claim (PD17 / LG): Team smooth L_C on real rosters vs LG sim — same $\sigma(\gamma(\hat{A}_j - \theta))$ formula; compare distribution shape before SCORE.

Empirical vs LG — team L_C distribution (normalized)

Same L_C formula; normalized so team-season count differences don't dominate the visual.

MBB 2011-2021 · team smooth L_C · LG $\rho = 0.5$ · $\gamma = 0.5$ · $\theta = 2.295z$ ($K/N = 0.0182$)

- Team $L_C = \text{mean}_j \sigma(\gamma(\hat{A}_j - \theta))$ — team-smooth congestion.
- Left panel: empirical NCAA (real rosters). Right: LG ASSIGN sim.
- Panel: MBB 2011-2021 · PPM z · min 20 min · poolq winsor 0.01–0.99.
- Shared $\theta = F_A^{-1}(1 - K/N) = 2.295z$ ($K/N = 0.0182$).
- $\gamma = 0.5$ (HAND17 slide-4 default; 539 placeholder).
- Empirical: mean $L_C = 0.254$, sd= 0.039 (6,492 team-seasons).
- LG ($\rho = 0.5$): mean $L_C = 0.251$, sd= 0.036 (4,140 team-seasons).
- Sim: one LG league per season ($J = N/15$), stacked team-seasons.
- ASSIGN shapes rosters; L_C is score-side input — computed after assign.

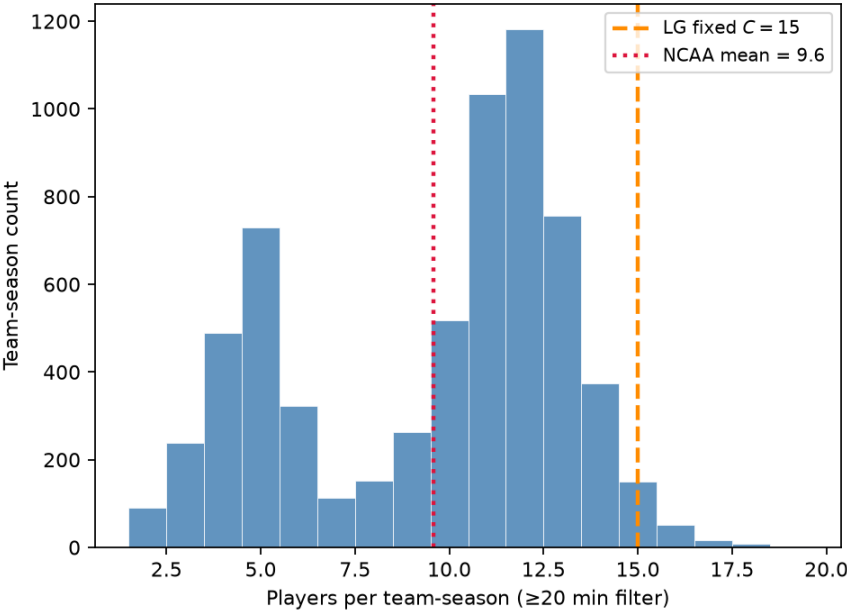


Claim (PD17 / LG): Team smooth L_C on real rosters vs LG sim — same $\sigma(\gamma(\hat{A}_j - \theta))$ formula; compare distribution shape before SCORE.

NCAA roster sizes after the minutes filter

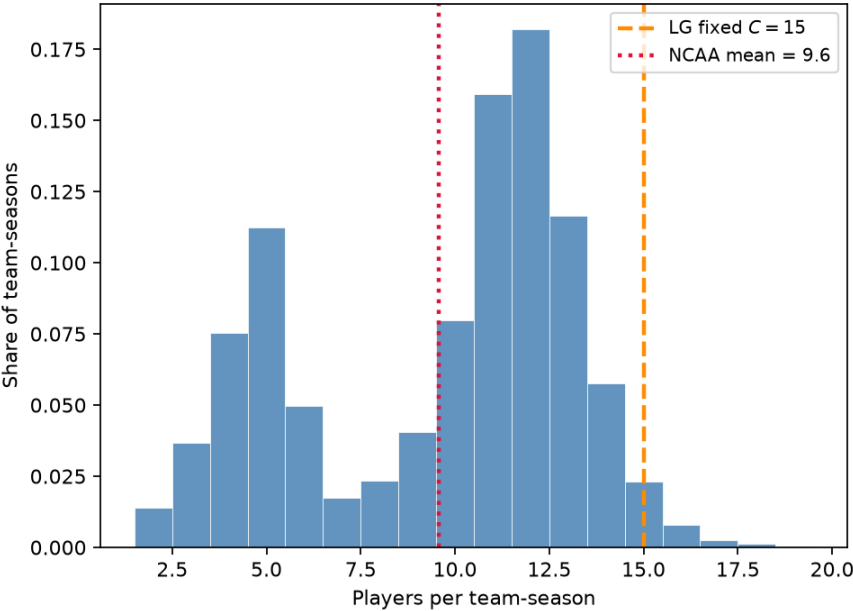
MBB 2011-2021 · min 20 min · mean = 9.6 vs LG C = 15

NCAA roster sizes fed to PD17 / LG analysis (2011-2021) — $n = 6,492$ team-seasons, mean=9.6, median=11, sd=3.6
NCAA roster sizes — team-season histogram



- Qualifying players per real NCAA team-season (≥ 20 ESPN minutes).
- $n = 6,492$ team-seasons · mean = 9.6 · median = 11 · sd = 3.6.
- Red dotted = NCAA mean; orange dashed = LG fixed $C = 15$.

Same distribution (normalized)



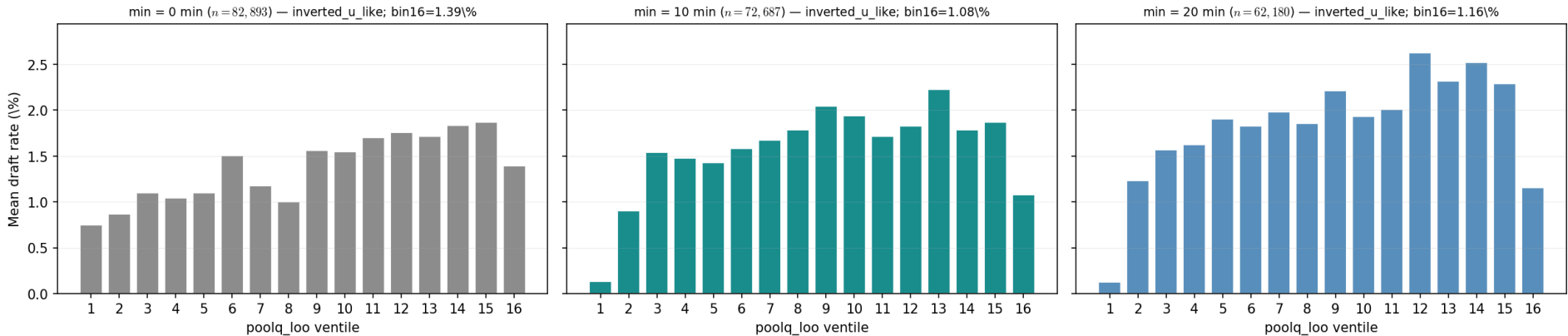
- Only 2.3% of teams have exactly 15 qualifying players.
- Same ability pool $\approx 62k$; LG repacks into $J = N/15$ synthetic 15-man leagues.
- Minutes floor shapes peer count — state explicitly in methods.

Talking point: empirical teams are ≈ 10 rotation players; LG uses fixed 15 — comparability is distributional, not headcount.

Hero sensitivity — min_minutes ladder

MBB 2011-2021 · poolq_loo ventiles · 0 / 10 / 20 min

Hero sensitivity — min_minutes ladder (MBB 2011-2021, 16 quantile, ppm z, winsor 0.01-0.99)

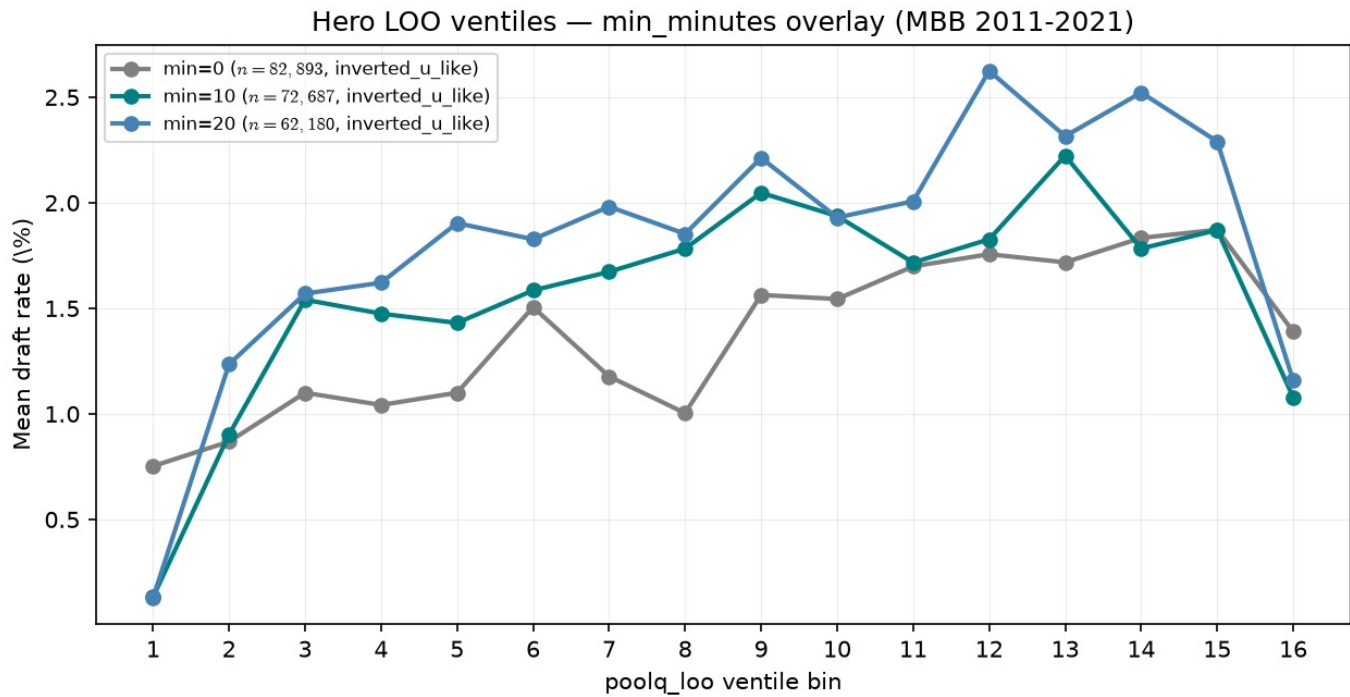


- Hero x-axis: draft rate vs poolq_loo ventiles · MBB 2011-2021 · 16 quantile.
- min_minutes defines who enters the panel and LOO peer pool — not neutral QC.
- min = 0: n = 82,893, drafted = 1,136, bin16 = 1.39%, inverted_u_like.
- min = 10: n = 72,687, drafted = 1,136, bin16 = 1.08%, inverted_u_like.
- min = 20: n = 62,180, drafted = 1,134, bin16 = 1.16%, inverted_u_like.
- Hero lock: min = 20 — cleanest top-ventile cliff; min = 10 robust appendix.
- We are not simulating minutes in LG.

Talking point: inverted-U survives 10- and 20-minute floors; min = 20 sharpens the elite ventile dip (+2 drafted vs min = 0).

Hero sensitivity — min_minutes overlay

MBB 2011-2021 · same bins, three floors



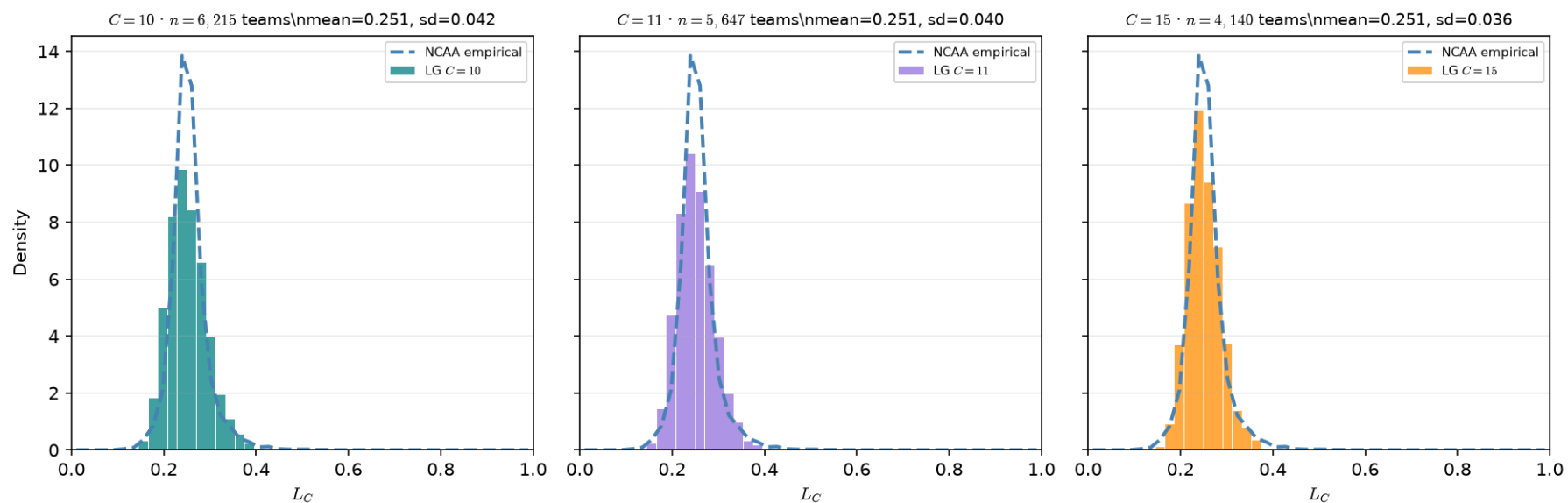
- Hero x-axis: draft rate vs poolq_loo ventiles · MBB 2011-2021 · 16 quantile.
- min_minutes defines who enters the panel and LOO peer pool — not neutral QC.
- min = 0: n = 82,893, drafted = 1,136, bin16 = 1.39%, inverted_u_like.
- min = 10: n = 72,687, drafted = 1,136, bin16 = 1.08%, inverted_u_like.
- min = 20: n = 62,180, drafted = 1,134, bin16 = 1.16%, inverted_u_like.
- Overlay: same estimand, different playing-time floors.
- We are not simulating minutes in LG.

Talking point: pattern is not an artifact of one arbitrary cutoff.

LG capacity sweep — team L_C vs NCAA

MBB 2011-2021 · $C \in 10, 11, 15$ · NCAA mean = 0.254

Team L_C — LG capacity sweep vs NCAA (MBB 2011-2021, $\rho = 0.5$, $\gamma = 0.5$)



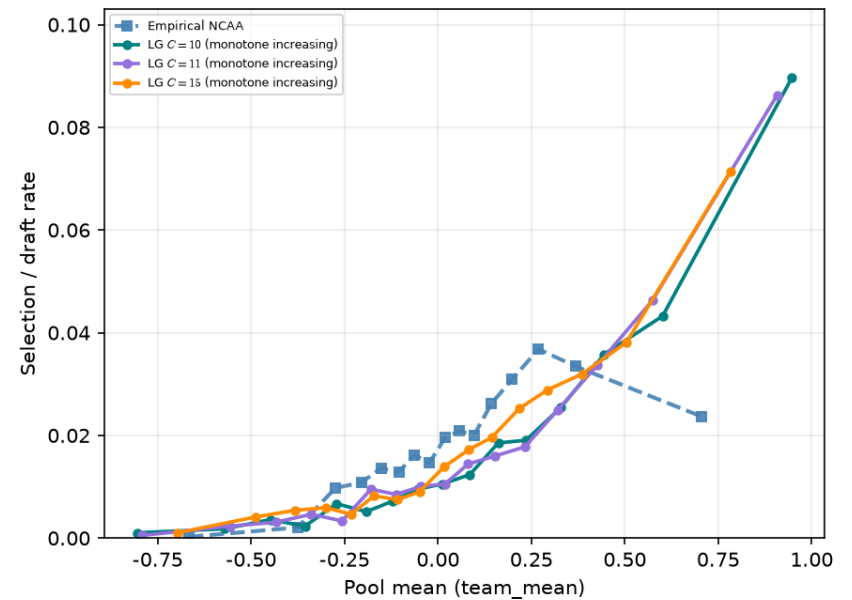
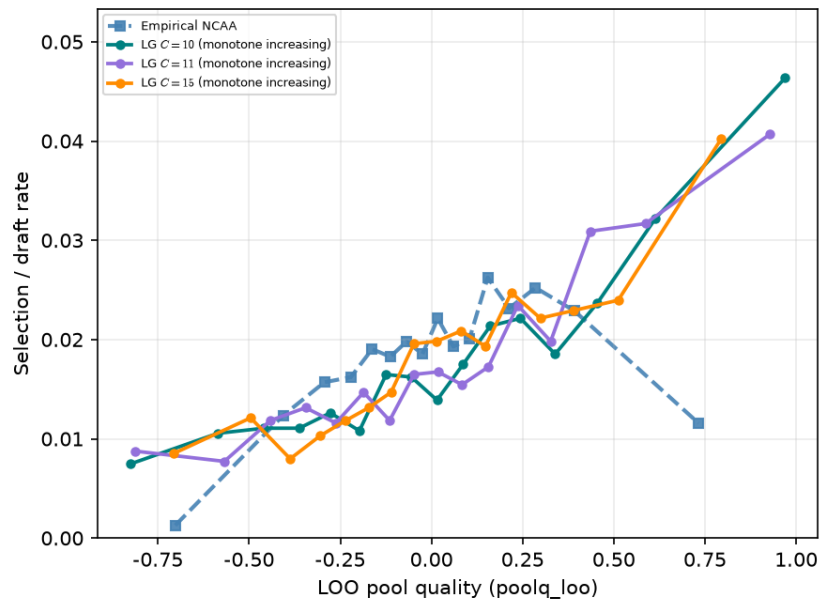
- Team L_C after ASSIGN — same $\sigma(\gamma(\hat{A}_j - \theta))$ as NCAA.
- MBB 2011-2021 · min 20 min panel · $\rho = 0.5$, $\gamma = 0.5$.
- NCAA: $n = 6,492$ team-seasons · mean = 0.254 · sd = 0.039.
- $C = 10$: $n = 6,215$ teams · mean = 0.251 · sd = 0.042.
- $C = 11$: $n = 5,647$ teams · mean = 0.251 · sd = 0.040.
- $C = 15$: $n = 4,140$ teams · mean = 0.251 · sd = 0.036.
- $C = 10$ closest to NCAA team count ($\approx 6.2k$ vs $6.5k$); $C = 15$ = gallery default.
- L_C first moment stable across capacity — score-side calibration OK.
- Capacity is a labeled sensitivity, not a silent swap for $C = 15$.

Talking point: $C = 10$ aligns team counts; L_C mean ≈ 0.251 at every C — capacity mismatch is interpretive, not a broken L_C calibration.

LG capacity sweep — SELECT vs empirical Hero

MBB 2011-2021 · LOO + pool mean · all LG curves monotone

LG SELECT — capacity C sweep vs empirical (MBB 2011-2021, $\rho = 0.5$)



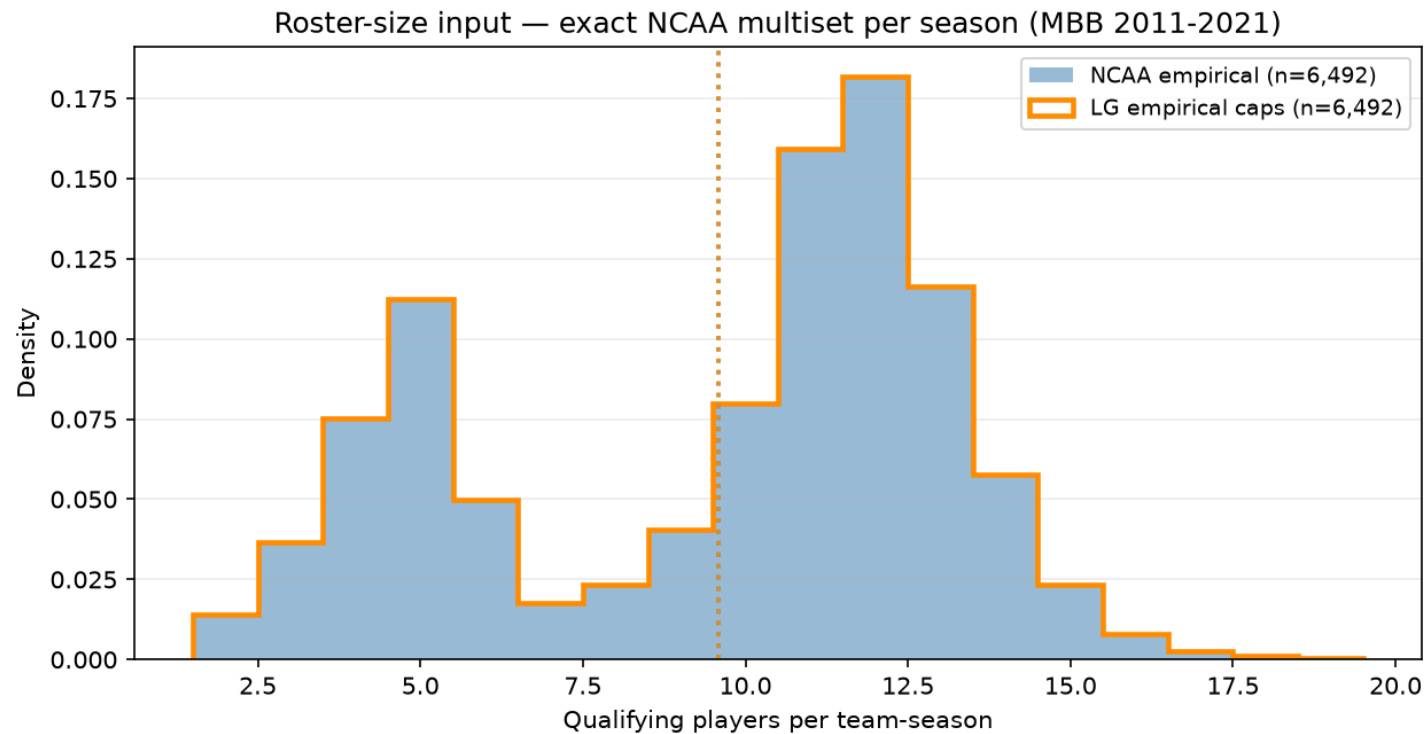
- SELECT: draft rate vs LOO peer quality (left) and pool mean (right).
- MBB 2011-2021 · fixed $\rho = 0.5$ · empirical dashed blue.
- Empirical LOO: inverted u like (peak bin 11,2.62%).
- LG $C = 10$: monotone increasing on LOO.
- LG $C = 11$: monotone increasing on LOO.

- LG $C = 15$: monotone increasing on LOO.
- Lowering C toward rotation-cohort scale does not recover inverted-U.
- ρ shapes ASSIGN sorting; selection curvature lives in SCORE (λ).
- λ sweep results → slides 20-21.

Talking point: C fixes headcount, not selection shape — empirical inverted-U survives; λ is the promising SCORE knob.

LG input — empirical roster-size multiset

MBB 2011-2021 · mean = 9.6 vs old fixed $C = 15$



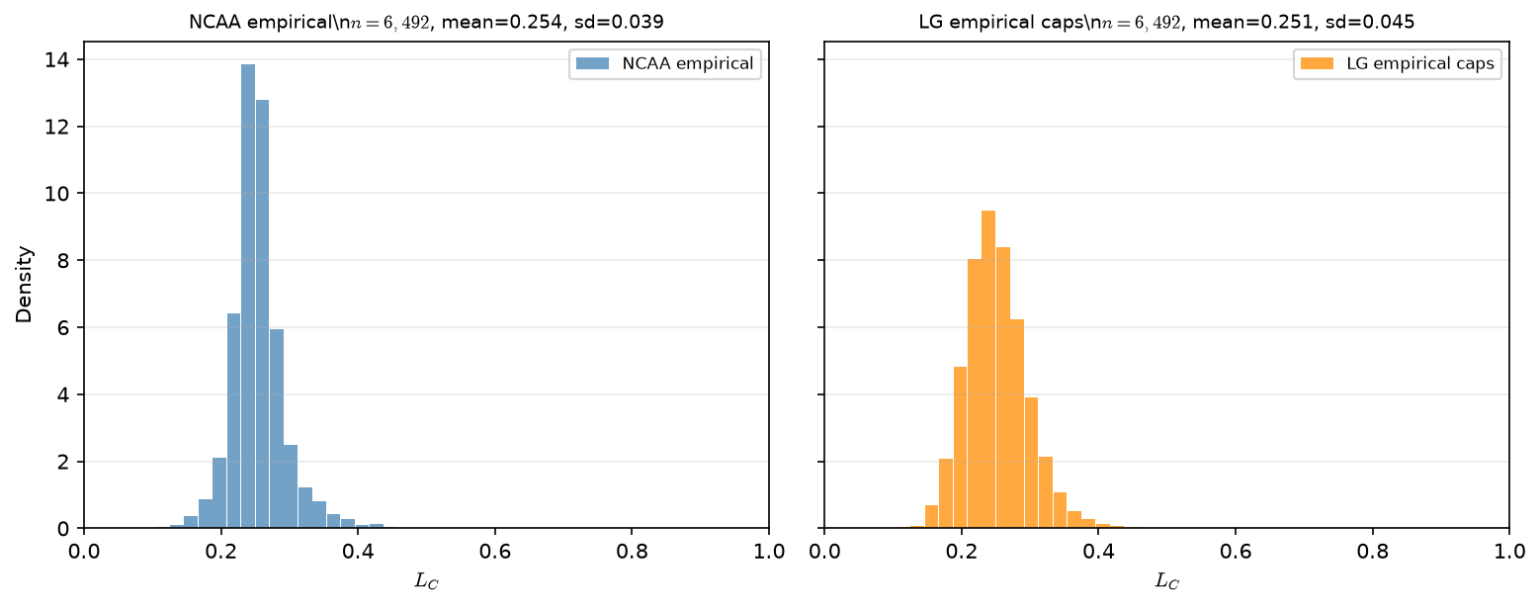
- Alex ask: LG stub capacities = exact NCAA filtered roster-size multiset.
- MBB 2011-2021 · min 20 min · one capacity per real team-season per year.
- NCAA / LG: $n = 6,492$ team-seasons · mean = 9.6 · median = 11 · sd = 3.6.
- Range 2– 19 qualifying players per team.
- Replaces fixed $C=15$ repack ($I = N/15$, every roster exactly 15).
- Same player pool ($\approx 62k$); ASSIGN homophily unchanged — only input geometry.
- LG realized roster sizes match NCAA by construction (overlay).

Talking point: Alex's fix — LG teams get the exact filtered NCAA roster-size distribution, not a synthetic 15-man league.

Empirical caps — team L_C vs NCAA

MBB 2011-2021 · 6,492 team-seasons · NCAA mean = 0.254

Team L_C — LG with empirical roster caps vs NCAA (MBB 2011-2021, $\rho = 0.5$, $\gamma = 0.5$)



- Team L_C after empirical-cap ASSIGN vs real NCAA rosters.
- MBB 2011-2021 · $\rho = 0.5$, $\gamma = 0.5$.
- NCAA: $n = 6,492$ · mean = 0.254 · sd = 0.039.
- LG empirical caps: $n = 6,492$ · mean = 0.251 · sd = 0.045.

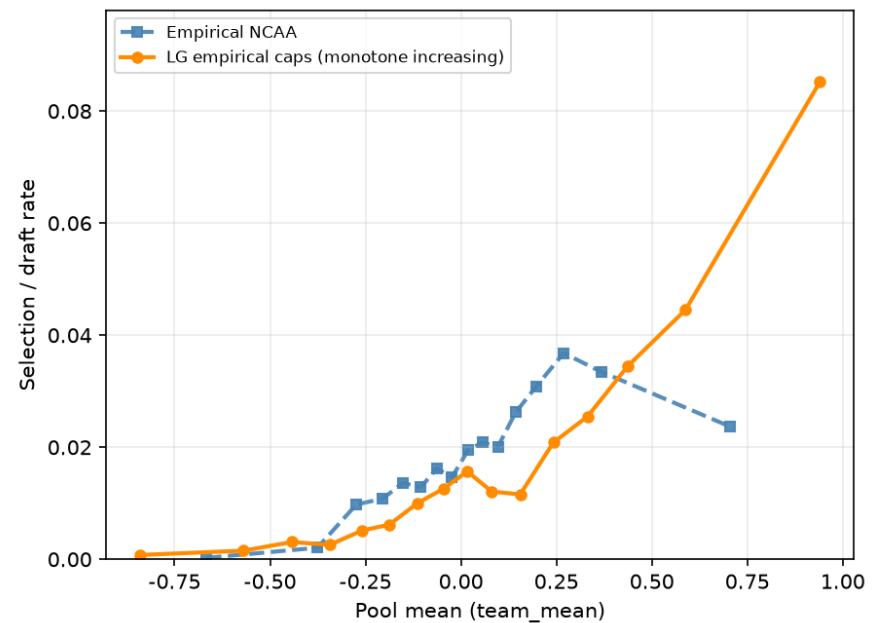
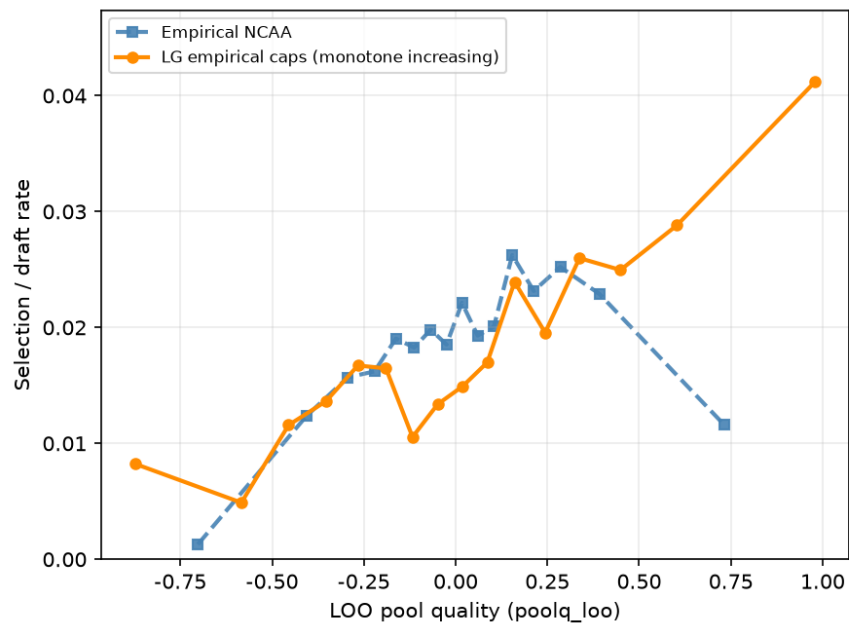
- Team count now aligned (6,492) — fixed $C = 15$ had only 4,140 teams.
- L_C first moment still matches; slightly wider LG tail (assign noise).
- Input comparability fixed; SCORE/SELECT shape is a separate question.

Talking point: comparability achieved — same team count and L_C mean; assignment layer is now apples-to-apples on roster geometry.

Empirical caps — SELECT vs Hero

MBB 2011-2021 · LOO still monotone · empirical inverted-U

LG SELECT — empirical roster caps vs NCAA (MBB 2011-2021, $\rho = 0.5$)



- SELECT: draft rate vs LOO peer quality (left) and pool mean (right).
- MBB 2011-2021 · empirical caps · $\rho = 0.5$ · default λ .
- Empirical LOO: inverted u like (peak bin 11, 2.62%).
- LG empirical caps: monotone increasing on LOO.

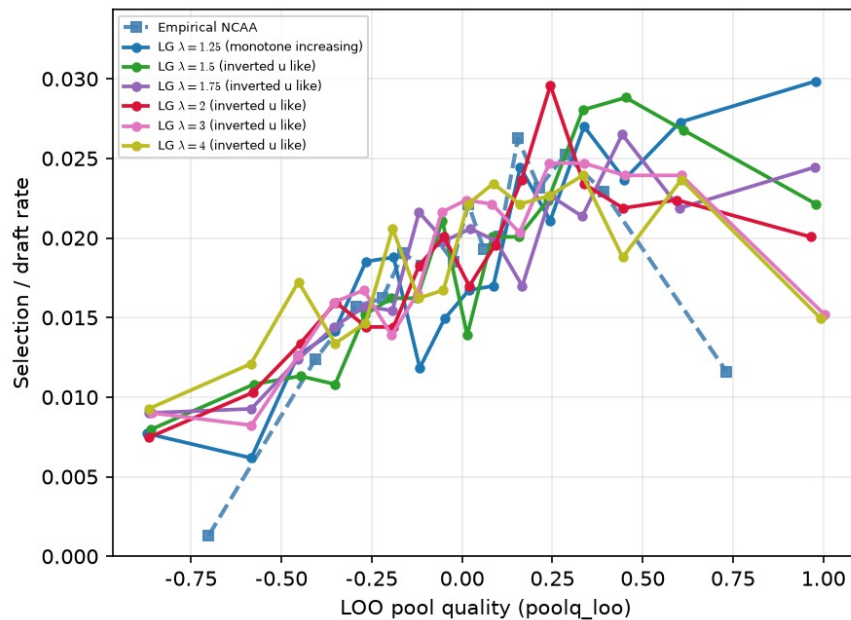
Talking point: input fix necessary but not sufficient — λ sweep on SELECT is the next experiment.

- Roster-size alignment does not recover Hero inverted-U alone.
- ρ = ASSIGN sorting; λ = SCORE congestion penalty — next sweep.
- C sweep and exact caps both monotone — shape gap is not headcount.

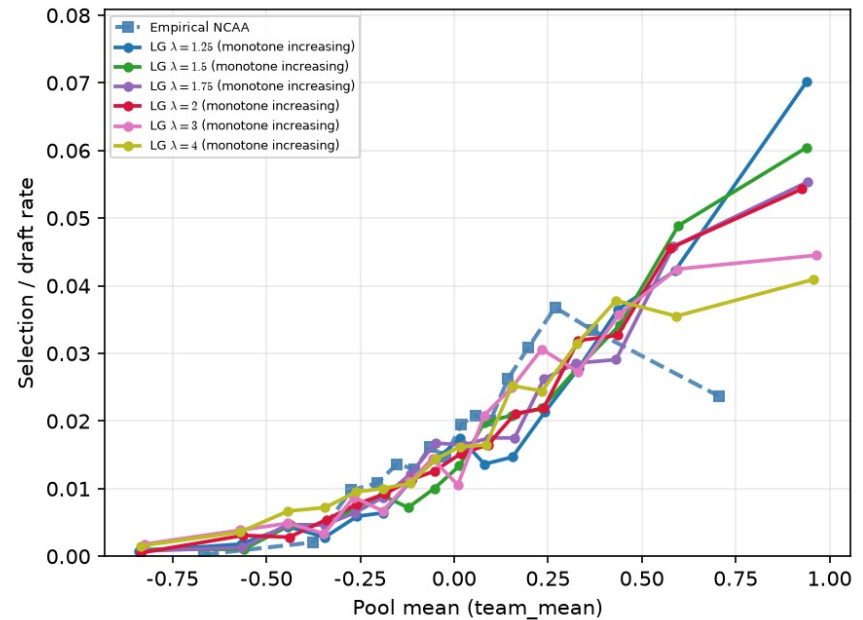
LG SELECT — λ sweep (fine grid)

MBB 2011-2021 · empirical roster caps · $\rho = 0.5$ · $S = A - \lambda C$ · $\lambda \in 1.25, 1.5, 1.75, 2, 3, 4$

LG SELECT — λ sweep vs empirical (MBB 2011-2021, empirical caps, $\rho = 0.5$, $S = A - \lambda L_C$)



- Empirical NCAA (dashed): inverted-U on LOO; peak bin 11 (~2.6% draft rate).
- $\lambda = 1.25$: still monotone $\uparrow$ on LOO — last arm before the flip.
- $\lambda = 1.5$: first inverted-U-like LOO curve.



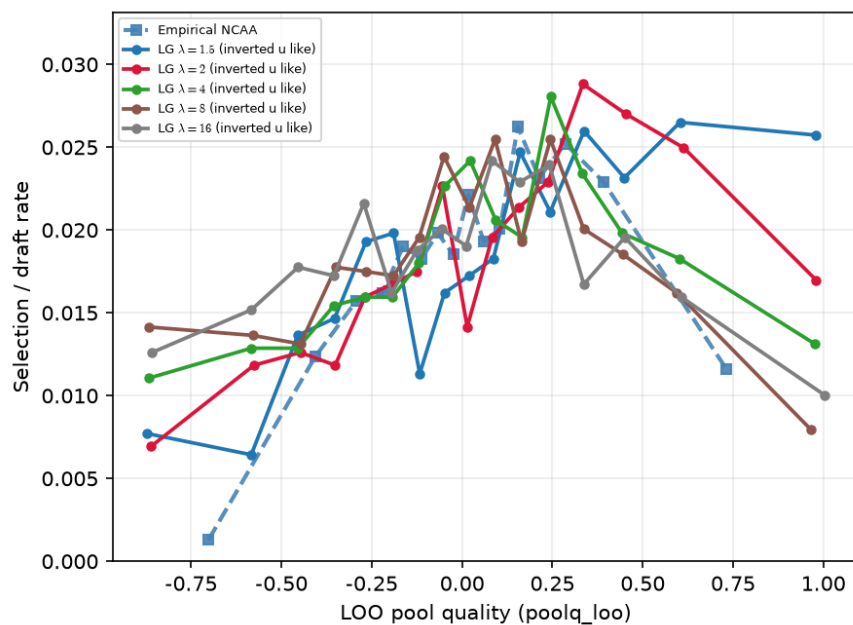
- $\lambda = 2, 3$: inverted-U on LOO; **peak bin 11 matches empirical**.
- Pool mean (right): all arms monotone $\uparrow$ at these λ — Hero on LOO axis only here.
- **Takeaway**: with inputs fixed, curvature lives in **SCORE** (λ), not ASSIGN (ρ) or caps.

Talking point: With NCAA-comparable inputs fixed, LOO SELECT stays monotone at $\lambda = 1.25$ and flips to inverted-U by $\lambda = 1.5$; $\lambda = 2-3$ matches the **empirical peak bin (11)** on LOO — curvature lives in **SCORE**, not ASSIGN.

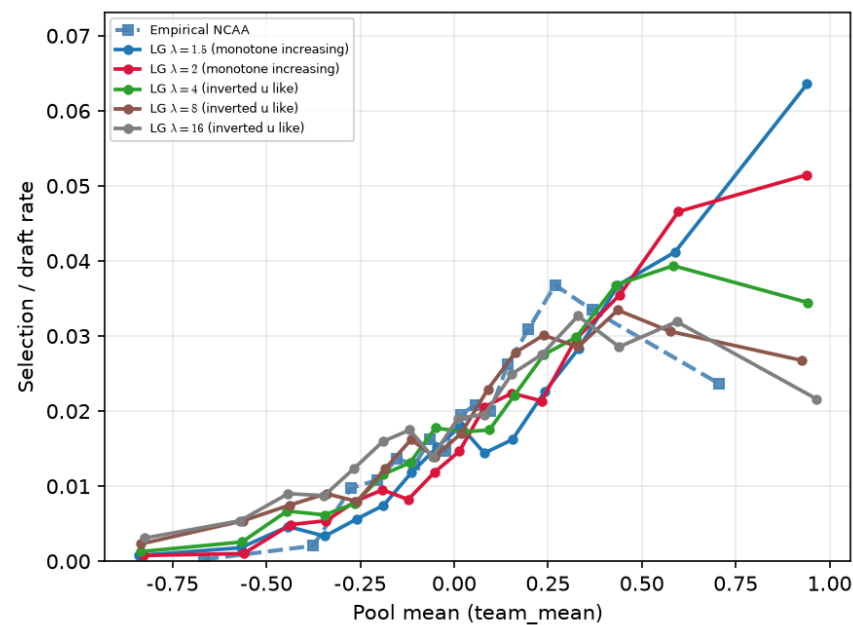
LG SELECT — λ sweep (extended arms)

Same panel · $\lambda \in \{1.5, 2, 4, 8, 16\}$

LG SELECT — λ sweep vs empirical (MBB 2011-2021, empirical caps, $\rho = 0.5$, $S = A - \lambda L_C$)



- LOO (left): inverted-U persists for **all** arms ≥ 1.5 .
- $\lambda = 2, 4$: peak bins 11–12 — closest to empirical shape on LOO.
- $\lambda = 8, 16$: peak shifts **left** (bins 9–12) — high λ over-penalizes elite pools.



- Pool mean (right): $\lambda = 4, 8, 16$ show inverted-U-like shape; $\lambda = 1.5, 2$ still monotone.
- **Takeaway:** λ controls both **whether** and **where** the dip appears; very high λ is not “more empirical.”

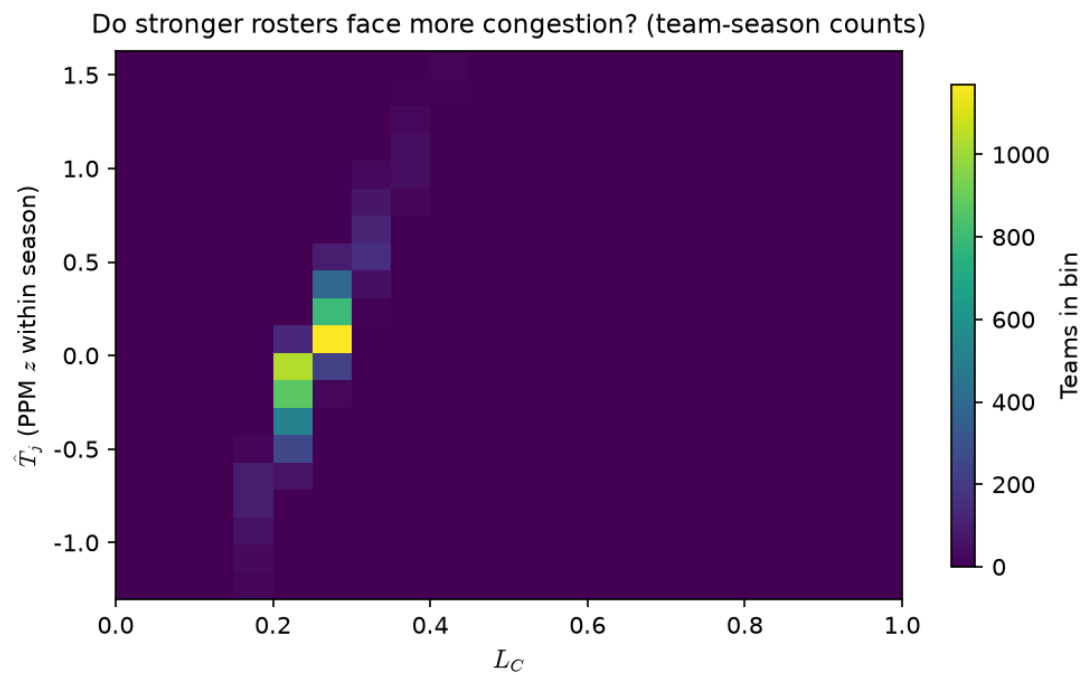
Talking point: Inverted-U on LOO **persists** for $\lambda \geq 1.5$, but $\lambda = 8\text{--}16$ shifts the peak **too far left**; on pool mean, inverted-U appears only at $\lambda \geq 4$ — so “more λ ” is not uniformly “more empirical.”

PD17 — Empirical Sketch A: $\hat{T}_j$ vs L_C

MBB 2011-2021 · team smooth L_C · PPM z within season · min 20 min · $\theta = F_A^{-1}(1 - K/N) = 2.295$. ($K/N = 0.0182$) · $\gamma = 0.5$ placeholder

- Each cell: count of team-seasons at $(\hat{T}_j, L_C)$.
- Color = count (empirical Sketch A — no ρ arms).
- $\hat{T}_j$ = mean $\hat{A}_i$ on roster (realized talent, not T_{j*}).
- Look for upward co-movement: higher $\hat{T}_j$ with higher L_C .
- Panel: mean $\hat{T}_j = 0.011$, mean $L_C = 0.254$.

Empirical Sketch A — $\hat{T}_j$ vs L_C ($\theta = 2.295$ z-units, $\gamma = 0.5$)

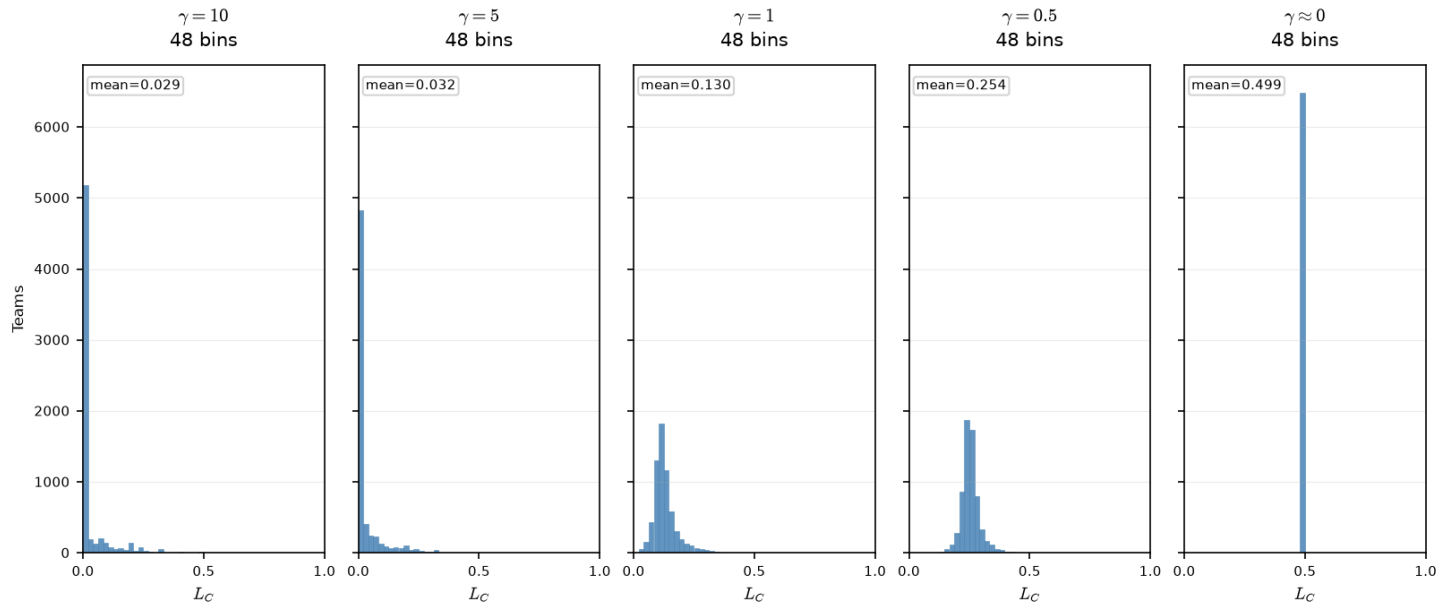


Claim (PD17): On actual team-seasons, stronger realized rosters ($\hat{T}_j$) tend to co-occur with higher L_C — empirical Sketch A.

PD17 — Empirical γ sweep: team L_C on real rosters

MBB 2011-2021 · team smooth L_C · PPM z · min 20 min · $\theta = 2.295$ z-units ($K/N = 0.0182$) fixed across γ arms

Empirical MBB — team L_C vs γ ($\theta = 2.295$ z-units, $K/N = 0.0182$)



OAT on real panel: fixed θ (K/N cutline on PPM z); only γ in $\sigma(\gamma(\hat{A} - \theta))$ changes.

Arms: $\gamma \in \{10, 5, 1, 0.5, 0.001\}$ — same team-seasons, same rosters.

At $\gamma = 10$: mean $L_C = 0.029$, 84% teams below 0.05.

At $\gamma \approx 0$: mean $L_C = 0.499$, max $L_C = 0.500$ — spread opens on z-scale.

Claim (PD17): On real rosters, γ reshapes team L_C — high γ compresses viability near 0 on the PPM-z scale; low γ spreads L_C into the interior of $[0,1]$.